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Roll No

EC-225-CBCS

B.E. IV Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Signals and Systems

Time : Three Hours

Maximum Marks : 60

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1.
 - a) Derive and explain the convolution integral.
 - b) Write the classification of signals and explain.
2.
 - a) Write about basic system properties.
 - b) Properties of convolution with proof.
3. Write a short note on:
 - a) Relationship between DTFT and Z-transform
 - b) Properties of the ROC
 - c) Properties of Z-transforms
4.
 - a) Properties of convolution summation?
 - b) Impulse response DT-LTI system and properties.
5.
 - a) Explain about the Fourier transform for Periodic signal.
 - b) Explain the representation of Aperiodic signal.

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6. Obtain the direct form-I, direct form-II cascade and parallel form realization of the LTI system governed by the equation.

$$y(n) = \frac{-13}{12} y(n-1) - \frac{9}{24} y(n-2) - \frac{1}{24} y(n-3) + x(n) + 4x(n-1) + 3x(n-2)$$

7. Write a short note on:
- a) Application of DTFS
 - b) Applications of DTFT
 - c) Unilateral Z transforms
 - d) Inverse Z transforms
8. a) Find $X(z)$ and sketch the zero dash pole plot and the ROC for $a < 1$ and $a > 1$ for the signal $X[n] = a^{|n|}$
- b) Determine the inverse z transform of

$$X(z) = \log\left(\frac{1}{1 - az^{-1}}\right)$$

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